



TB AUDIO PLUG-INS
Team Impulse Impact

USER MANUAL

TB Disperser

Phase Dispersion

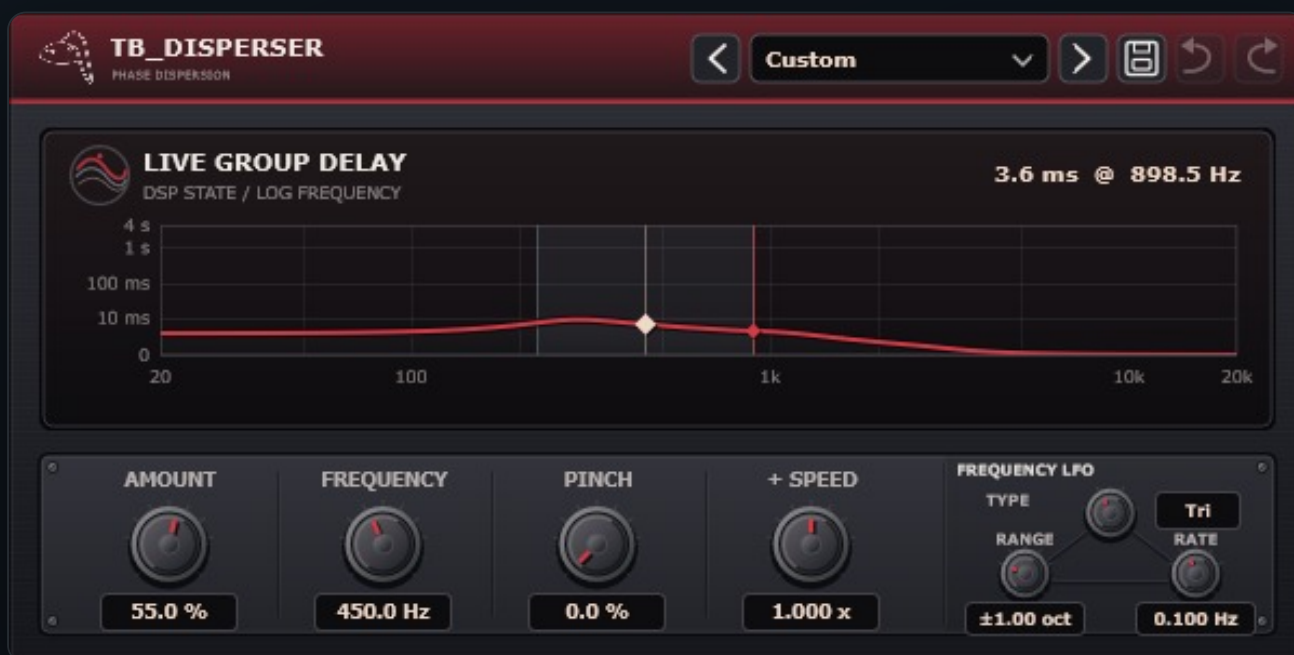
All-pass phase dispersion that stretches a sharp transient into a heavier, frequency-dependent sweep, with a built-in frequency LFO for motion.

All-pass dispersion

Live group-delay display

Frequency LFO

10 factory presets



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01 Overview

TB Disperser delays different frequencies by different amounts. A drum hit or synth attack, which normally arrives all at once, is spread into a short sweep, the familiar laser-like zap of phase dispersion.

Because it changes when frequencies arrive rather than how loud they are, TB Disperser can make drums, bass, and synth attacks feel longer, heavier, and more animated without adding distortion. A dedicated LFO moves the dispersion frequency for motion without automation.

Key features

- Chain of all-pass sections whose group delay is concentrated around a chosen frequency
- AMOUNT, FREQUENCY, PINCH, and SPEED to shape the length and focus of the sweep
- Frequency LFO with five shapes, a range of up to ± 4 octaves, and a free-running rate
- Live display of the group delay across the spectrum, with a readout at the current LFO frequency
- Ten factory presets with descriptions, plus user presets, undo, and redo

Signal flow



About this manual

This manual describes TB Disperser 1.3.0 for Windows (VST3). The current macOS build is version 0.1.2 and may differ from the descriptions here.

02 Quick start

- 1 Insert TB Disperser on a drum, bass, or synth track. The defaults are **AMOUNT** 70%, **FREQUENCY** 200 Hz, **PINCH** 0%, **+ SPEED** 2.0x, with the LFO off.
- 2 Play the material and move **FREQUENCY** **5** to the area where you want the sweep: low values for weight, higher values for a brighter zap.
- 3 Raise **AMOUNT** **4** for a stronger, longer dispersion.

- 4 Lower **+ SPEED** 7 to stretch the zap into a longer, slower sweep.
- 5 Raise **PINCH** 6 to concentrate the ringing around FREQUENCY.
- 6 For motion, choose an LFO **TYPE** 8 and set **RANGE** and **RATE**.

03 Interface



- 1 **Preset bar** — previous and next preset (hold to browse), the Factory and User preset list, Save User Preset, Undo, and Redo.
- 2 **LIVE GROUP DELAY** — how much each frequency is delayed, on a log-frequency scale.
- 3 **Live readout** — the group delay at the frequency currently produced by the LFO.
- 4 **AMOUNT** — how much phase dispersion accumulates.
- 5 **FREQUENCY** — where the group delay is concentrated.
- 6 **PINCH** — how tightly the ringing gathers around FREQUENCY.
- 7 **+ SPEED** — the speed of the dispersion; lower values give a longer zap.
- 8 **FREQUENCY LFO** — **TYPE**, **RANGE**, and **RATE** of the frequency modulation.

04 Controls

Phase dispersion

Control	Range · default	What it does
AMOUNT	0 to 100% · 70%	Sets the amount of accumulated phase dispersion by setting how many all-pass sections are active. Higher values give a stronger, longer sweep.
FREQUENCY	20 Hz to 20 kHz · 200 Hz	Sets the frequency where group delay is concentrated. Low settings add weight; high settings give a bright, metallic zap.

Control	Range · default	What it does
PINCH	0 to 100% · 0%	Concentrates ringing around the selected frequency. At 0% the sections spread over up to ± 2 octaves; at 100% they converge into a narrow, ringing focus.
+ SPEED	0.25 to 2.0x · 2.0x	Sets the dispersion speed. Lower values create a longer, slower zap.

Frequency LFO

Control	Range · default	What it does
TYPE	Off, Sine, Tri, Saw, Sqr, Rnd · Off	Turns the LFO off or selects one of five shapes. It is a stepped rotary control.
RANGE	0 to 4 oct (shown as $\pm$) · ± 1 oct	Frequency sweep depth in octaves around FREQUENCY. Zero turns modulation off.
RATE	0.01 to 10 Hz · 0.25 Hz	The free-running LFO rate in hertz. It does not follow the host tempo.

TB Disperser has no dry/wet control. All-pass dispersion changes timing, not level, but if you blend its output with the dry signal on a parallel bus, phase differences can cancel parts of the sound. Check the result in mono.

05 Reading the display

LIVE GROUP DELAY plots how long each frequency is delayed (vertical axis, from 0 up to seconds) across the spectrum (horizontal axis, 20 Hz to 20 kHz). The peak of the curve sits around FREQUENCY; AMOUNT raises it, PINCH narrows it, and lower SPEED values stretch it.

With the LFO running, the marker moves with the modulated frequency, and the readout at the top right shows the group delay at that frequency, for example 3.6 ms @ 898.5 Hz.

06 Presets

Use the arrows **1** to step through presets (hold to browse quickly), or open the list to choose a Factory or User preset; Space also opens it. Save User Preset stores the current settings under a name of your choice. Undo and Redo step through parameter and preset changes.

Preset	Group	Description
Init	Essentials	A safe, already audible starting point for phase dispersion.
Tight Phase Shape	Essentials	Adds a short, light upper-mid phase shape to percussion and synth attacks.
Drum Bloom	Essentials	Opens drum transients with a practical medium-length bloom.
Slow Drum Zap	Essentials	Stretches drum attacks into a slow zap.
Slow Metallic Bloom	Essentials	Uses focused ringing and the slowest dispersion for a metallic showcase.

Preset	Group	Description
Bass Bloom	Motion	Breathes slowly around the bass fundamental without a wide sweep.
Synth Drift	Motion	Moves a sustained synth smoothly through a broad ten-second triangle cycle.
Rising Zap	Motion	Creates a directional rising zap that resets for drums and synths.
Two Point Pulse	Motion	Alternates clearly between two phase centres for a stepped rhythmic pulse.
Random Scatter	Motion	Holds unpredictable phase centres long enough to make each scatter intentional.

07 Recipes

Starting points for common jobs. Adjust them by ear.

Laser snare or clap

A bright, obvious zap on short percussion.

Preset	Slow Drum Zap
FREQUENCY	1 to 3 kHz
+ SPEED	Lower for a longer zap

- 1 Raise AMOUNT until the zap is clear, then back off so the hit stays tight.

Heavier kick or bass attack

More weight and length without distortion.

Preset	Bass Bloom
FREQUENCY	50 to 150 Hz
AMOUNT	Start below 70%

- 1 Low FREQUENCY settings can sound like a wide phase shift; keep AMOUNT moderate.

Moving synth pad

Slow, evolving phase motion on sustained sounds.

Preset	Synth Drift
TYPE	Tri or Sine
RATE	0.05 to 0.2 Hz
RANGE	±1 to ±2 oct

- 1 Start with a small RANGE and slow RATE, then widen.

Rhythmic phase pulse

A stepped, rhythmic change of colour.

Preset	Two Point Pulse
TYPE	Sqr
RATE	Set by ear against the groove

- 1 RATE is free-running; adjust it until the steps sit with the music.

08 Tips and troubleshooting

Symptom	What to check
The low end sounds smeared or hollow.	Lower AMOUNT or raise FREQUENCY. Low frequencies are delayed the most.
The effect is too subtle.	Raise AMOUNT, lower + SPEED, or raise PINCH for a more focused ring.
The sound rings too much.	Lower PINCH so the dispersion spreads over a wider range.

Symptom	What to check
The LFO does nothing.	Choose a TYPE other than Off and set RANGE above zero.
The LFO is not in time with the song.	RATE is free-running in hertz; adjust it by ear or automate it.
Parts of the sound cancel in mono.	This happens when the dispersed signal is blended with the dry one. Reduce AMOUNT or avoid parallel blending.

09 Parameter reference

Panel label	Host parameter	Range	Default
AMOUNT	Amount	0 to 100%	70%
FREQUENCY	Frequency	20 Hz to 20 kHz	200 Hz
PINCH	Pinch	0 to 100%	0%
+ SPEED	Speed	0.25 to 2.0x	2.0x
LFO TYPE	LFO Type	Off, Sine, Tri, Saw, Sqr, Rnd	Off
LFO RANGE	LFO Range	0 to 4 oct	1 oct
LFO RATE	LFO Rate	0.01 to 10 Hz	0.25 Hz

10 Version history

Version	Changes
1.3.0 · 2026-07-30	The plug-in logo and name share one rack header. Previous/next navigation and save/load in the preset control were improved.

11 Support and license

Report problems or request features at github.com/TaebeumKim/TB_Audio-Plugins/issues. Include the plug-in version, your DAW, and your operating system.

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